

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1064

Casey West 100 kV
January 29, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1064. The Interconnection Customer proposes a 44.9 MW solar photovoltaic facility interconnecting at the Casey West 100 kV substation in the Illinois area. The request was received on September 13, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 380 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,593,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mouton - Ledges 230kV	Ravenna - El Paso 500kV L	107.9	675	5.7
Waynedale - Montpelier Electric Generating Station Substation 230kV	Richland - Dekalb 500kV	107.6	1,266	20.7
Sidney A Murray Jr Hydroelectric - 24Th & Gardendale 230kV	Dekalb - Richland 500kV	107.1	860	9.1
Troy Energy Llc - Richland 500kV	Swan Vly - Trego 500kV L	106.6	4,387	37.2
Winthrop - Hebo 230kV Line	Barrett - Hebo 230kV L	105.9	1,152	5.5
Twin County Ranch - Richland 230kV	Troy Energy Llc - Richland Substation 500kV	104.2	1,454	19.2
Leetonia - Maple St 230kV	Ravenna - El Paso 500kV line	103.8	829	5.1
Atascocita - Maple St 230kV	El Paso - Ravenna 500kV	103.7	848	5.1
Moton - Treat 230kV	El Paso - Ravenna 500kV L	103.4	722	5.7
Muskogee - Darlington Rd 100kV Line	Richland - Dekalb 500kV line	103.4	658	6.6
Richland 500/230 kV Transformer Bank	Troy Energy Llc - Richland 500kV	103.3	2,672	32.2
Gavin Power Llc 500-345 kV TX	Olympus 500/230kV Auto	103.3	1,310	5.1
Ravenna - Old Forge 230kV	Hylebos - Ravenna 500kV	103.0	1,367	8.3
Trego - Beaver 500kV	Adams - Onancock Iesel Generation Facility Substation 230kV line	103.0	6,220	30.5
Montpelier Electric Generating Station - Musogee 500kV	Sabanera - Lonetree Substation 500kV Line	102.8	4,673	10.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Lookingglass - Corridor 230kV	Beaver - Trego 500kV	102.8	1,503	9.8
Dekalb Substation - Robiso Park Substation 230kV	Jones Chapel - Dekalb 500kV	102.6	3,311	33.7
Lookingglass - Truman 230kV	Trego - Beaver 500kV	102.4	1,538	9.8
Pickford - Adams 230kV line	Trego - Beaver 500kV Line	102.2	1,595	8.0
Waldo - Truman 230kV	Trego - Beaver 500kV	102.1	1,568	9.8
Onancock Diesel Generation Facility - Adams 230kV	Trego - Beaver 500kV	101.4	3,126	15.7
Sta S Van Nuys Substation 500/230kV Transformer Bank	Troy Energy Llc - Richland 500kV	101.3	871	7.8
Dekab - Richland 500kV	Swan Vly - Trego 500kV L	101.3	4,730	49.3
State Street - Waynedale 230kV L	Dekalb - Richland 500kV	101.0	991	17.5
Monfort - Wabash 230kV	Dekalb - Richland 500kV	100.9	1,592	18.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mouton - Ledges 230kV	Reconductor line	\$6,134,000	0% (\$14,000)
Waynedale - Montpelier Electric Generating Station Substation 230kV	Reconductor line	\$17,462,000	0% (\$45,000)
Sidney A Murray Jr Hydroelectric - 24Th & Gardendale 230kV	Reconductor line	\$13,669,000	0% (\$38,000)
Troy Energy Llc - Richland 500kV	Reconductor line	\$65,315,000	0% (\$173,000)
Hebo - Winthrop Substation 230kV	Reconductor line	\$3,858,000	0% (\$8,000)
Twin County Ranch - Richland Substation 230kV L	Reconductor line	\$14,098,000	0% (\$40,000)
Maple St - Leetonia 230kV	Reconductor line	\$6,693,000	0% (\$17,000)
Atascocita - Maple St 230kV	Reconductor line	\$10,804,000	0% (\$27,000)
Moton - Treat 230kV	Reconductor line	\$9,751,000	0% (\$23,000)
Muskogee - Darlington Rd 100kV Line	Reconductor line	\$10,269,000	0% (\$28,000)
Richland 500/230 kV Transformer Bank	Replace transformer with higher-rated unit	\$41,053,000	0% (\$116,000)
Gavin Power Llc 500-345 kV XFMR	Replace transformer with higher-rated unit	\$33,583,000	0% (\$76,000)
Ravenna - Old Forge 230kV	Reconductor line	\$7,162,000	0% (\$17,000)
Trego - Beaver 500kV	Reconductor line	\$116,537,000	0% (\$279,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Muskogee - Montpelier Electric Generating Sta 500kV	Reconductor line	\$54,467,000	0% (\$88,000)
Lookingglass - Corridor 230kV	Reconductor line	\$6,580,000	0% (\$16,000)
Dekalb - Robison Park 230kV	Reconductor line	\$13,796,000	0% (\$38,000)
Lookingglass - Truman 230kV	Reconductor line	\$9,978,000	0% (\$25,000)
Pickford - Adams 230kV line	Reconductor line	\$8,324,000	0% (\$20,000)
Waldo - Truman 230kV	Reconductor line	\$8,641,000	0% (\$21,000)
Onancock Diesel Generation Facility - Adams 230kV	Reconductor line	\$23,378,000	0% (\$57,000)
Sta S Van Nuys 500-230 kV Transformer Bank	Replace transformer with higher-rated unit	\$46,144,000	0% (\$129,000)
Richland - Dekalb 500kV line	Reconductor line	\$92,883,000	0% (\$258,000)
Waynedale - State St 230kV	Reconductor line	\$5,256,000	0% (\$14,000)
Monfort - Wabash 230kV	Reconductor line	\$10,015,000	0% (\$26,000)
	Total allocated to project		\$1,593,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.